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A physics-informed machine learning framework for understanding the rejection mechanism of hydrophilic micropollutants in polyamide membranes

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ABSTRACT

Polyamide membrane-based separation technology has been widely applied for removing organic micropollutant from aqueous environments. However, the high solubility and diverse substructures of hydrophilic micropollutants (HMPs) have resulted in limited removal efficiency by polyamide membranes, and the removal mechanism has not been adequately explained by conventional steric and electrostatic models. To gain deeper mechanism understanding, we developed a physics-informed machine learning (PIML) on machine learning-based artificial intelligence framework that integrates molecular fingerprints, physical models, and multilevel descriptors (including pollutant properties, membrane characteristics, and operating conditions). This framework provides a more comprehensive understanding of the removal mechanism and provides molecular-level insights into the rejection of HMPs by polyamide membranes. SHAP analysis via PIML identified that the three primary determinants of HMP rejection as: (1) solute (HMPs) – membrane interaction energies (ΔGs–m), where more negative values significantly improved rejection; (2) presence of hydroxyl (–OH) groups, which further enhanced removal efficiency; and (3) steric hindrance effects that modulated rejection under specific conditions. Moreover, comparative analysis of membranes with varying degree of crosslinking (DC) demonstrated that increased surface carboxyl density enhances binding affinity, indicating that rejection efficiency depends synergistically on interaction energetics and accessible functional groups. Molecular dynamics simulations supported these machine learning-derived insights. Through the integration of data-driven prediction, physical models, and molecular-scale mechanisms, PIML provides mechanistic insights and design guidelines focused on optimizing solute–membrane interactions. This is achieved by precisely controlling surface chemical functional groups and the DC in polyamide membranes to enable efficient removal of HMPs.

1. Introduction

The proliferation of organic micropollutants in aqueous environment systems has become a pressing concern, given their persistent nature, biological activity, and resistance to conventional treatment technologies [1,2]. These approaches often suffer from incomplete degradation, secondary pollution, or high operational costs [3,4]. In this context, membrane-based technologies have emerged as particularly attractive

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alternatives. Notably, nanofiltration (NF) and reverse osmosis (RO) technologies combine scalability, energy efficiency, and tunable separation selectivity, providing a robust and adaptable platform for the effective removal of organic micropollutants [5,6].

Among these, hydrophobic micropollutants have been extensively investigated and are generally removed more effectively due to their larger molecular sizes and stronger adsorption to polyamide membranes [7]. In contrast, hydrophilic micropollutants (HMPs), despite their low affinity to polyamide membranes, which theoretically facilitates rejection, often prove more challenging to remove in practice. This can be attributed to their high aqueous solubility, low molecular weights (and correspondingly small size), and weak sorption tendencies, as exemplified by caffeine (MW = 194.19 g/mol) and acetaminophen (MW = 151.16 g/mol) [8]. Furthermore, HMPs frequently exhibit high polarity and diverse substructures (e.g., hydroxyl, carboxyl, or amino functional groups), which collectively complicate their removal [9,10]. These features not only reduce the effectiveness of size-exclusion mechanisms but also restrict specific solute–membrane interactions, allowing many HMPs to permeate through membranes with little resistance [11,12]. Conventional rejection mechanisms based primarily on steric hindrance or electrostatic repulsion provide only partial explanation for HMPs rejection. Thus, to design membranes with superior HMPs rejection, it's essential to uncover additional separation mechanisms and establish quantitative relationship governing their removal.

Recent studies have leveraged machine learning (ML) to elucidate membrane separation mechanisms and predict micropollutant rejection with improved accuracy and interpretability. Lee et al. developed fingerprint-based ML models for RO/NF membranes, achieving high accuracy and SHAP-derived molecular insights, yet their approach considered only solute structures without membrane or operational factors [13]. Ignacz et al. established a hybrid mechanistic–data-driven framework for nanofiltration prediction and energy analysis [14], which emphasized process-level sustainability assessment. Jafari et al. integrated molecular, membrane, and operational descriptors to predict rejection in forward osmosis using ANN and GBR models [15]. However, the physical coupling between data features and interaction mechanisms was absent. Ignacz et al. applied graph neural networks to capture solute–solvent interactions in organic solvent nanofiltration [16], providing atom-level interpretability but lacking integration of membrane descriptors. Our group previously applied ML to elucidate the multiscale structural and operational factors governing ion selectivity in PA nanofiltration membranes [17], and further supported that membrane surface charge modulates rejection by interacting with localized charged groups of sulfamethoxazole via combined ML and molecular dynamics (MD) simulations [18]. Despite these advances, several challenges persist. Most current ML models rely on single-domain descriptors or are tailored to specific contaminants, which constrains their generalizability across diverse HMPs. Although our recent study [19] provided a broad evaluation across multiple micropollutant classes, it primarily identified general rejection trends rather than elucidating molecular-level mechanism specific to HPMs. In the present work we focus exclusively on HMPs and develop a physics-informed machine learning (PIML) framework that integrates molecular structural fingerprints, mechanistic physical descriptors, and process conditions to predict and interpret HMP rejection. By embedding mechanistic knowledge into the learning process, the PIML framework provides quantitatively accurate predictions while offering interpretable, mechanism-level insights that can guide polyamide membrane design.

The PIML data framework integrates three key modules: (i) molecular fingerprints encoding functional groups and structural motifs; (ii) physicochemical interaction parameters, including solute-membrane interaction energies (ΔGs-m) based on the extended derjaguin–landau–verwey–overbeek (XDLVO) theory [20], steric (Φ_S) and Donnan partitioning (Φ_D) terms derived from the Donnan Steric Pore Model with Dielectric Exclusion (DSPM-DE) model [21]; and (iii) supplementary descriptors encompassing micropollutant properties, membrane characteristics, and operating conditions. As illustrated in Fig. 1, this multi-scale descriptor system establishes an interaction knowledge framework to guide the design of HMPs-removal membranes.

To evaluate modeling performance, six machine learning algorithms—Random Forest (RF), Extreme Gradient Boosting (XGBoost), Light Gradient Boosting Machine (LightGBM), convolutional neural network (CNN), Transformer, and artificial neural network (ANN)—were employed [17,18,22]. Using the best-performing model, SHapley Additive exPlanations (SHAP) were applied to interpret feature importance, identifying two dominant factors governing solute rejection: ΔGs-m and the presence of hydroxyl groups (–OH) in HMPs. Partial dependence plot (PDP) was further used to quantify the effect of these key features on rejection behavior. MD simulations were subsequently conducted to validate the mechanistic interpretations derived from SHAP. This study proposes a predictive and interpretable framework that not only accurately estimates HMPs rejection, but also provides actionable insights for rational membrane design through tailored control of surface chemical functional groups and the degree of crosslinking (DC) in polyamide membranes.

2. Methods

2.1. Dataset collection and preprocessing

To construct a comprehensive dataset suitable for interaction-aware modeling of HMPs, we compiled a total of 353 data entries describing the rejection behavior of HMPs under a wide range of polyamide membrane separation conditions. HMPs were defined as neutral compounds with low hydrophobicity (log Kow <2), reflecting their high aqueous solubility and limited electrostatic interactions with membranes. The complete dataset was provided as a supplementary file (MembraneDataset_353.xlsx), where the last column ("Reference") specifies the corresponding literature source for each data entry. Each entry incorporated descriptors spanning three core domains: micropollutant physicochemical properties, membrane structural parameters, and key operational conditions. In total, 35 feature descriptors were compiled to represent the multidimensional system behavior (Supplementary Table 1). This curated dataset provided a robust foundation for subsequent machine learning model development and mechanistic interpretation of HMP rejection.

To enable more detailed structural representation and molecular-level analysis, Simplified Molecular Input Line Entry System (SMILES) strings were curated for each micropollutant molecule, either extracted directly from chemical databases (e.g., PubChem) or generated manually when necessary. In addition to empirical descriptors, we further incorporated physics-based model parameters to capture membrane–solute transport behaviors, including the $\Delta G_{s-m} = 2\left(\sqrt{\gamma_l^{LW}} - \sqrt{\gamma_s^{LW}}\right)\left(\sqrt{\gamma_m^{LW}} - \sqrt{\gamma_l^{LW}}\right)$, $\Phi_s = \left(1 - \frac{r_s}{r_p}\right)^2$, and $\Phi_D = \exp\left(\frac{-z_m eF}{k_B T}\Delta\psi_m\right)$ [20,21]. The detailed computational procedure for ΔGs–m was provided in the Supporting Information (Detailed calculation of ΔGs-m part), where the theoretical assumptions underlying the XDLVO are also explicitly stated.

2.2. Substructure selection and redundancy elimination

To enhance the interpretability and mechanistic relevance of molecular features, we first conducted a systematic screening of common substructures derived from SMILES representations. Specifically, we curated a library of over 50 frequently encountered functional groups (e.g., hydroxyl, amine, sulfonyl, aromatic rings), and identified the 12 most prevalent ones across the dataset of HMPs. These representative substructures were subsequently incorporated into our feature set for

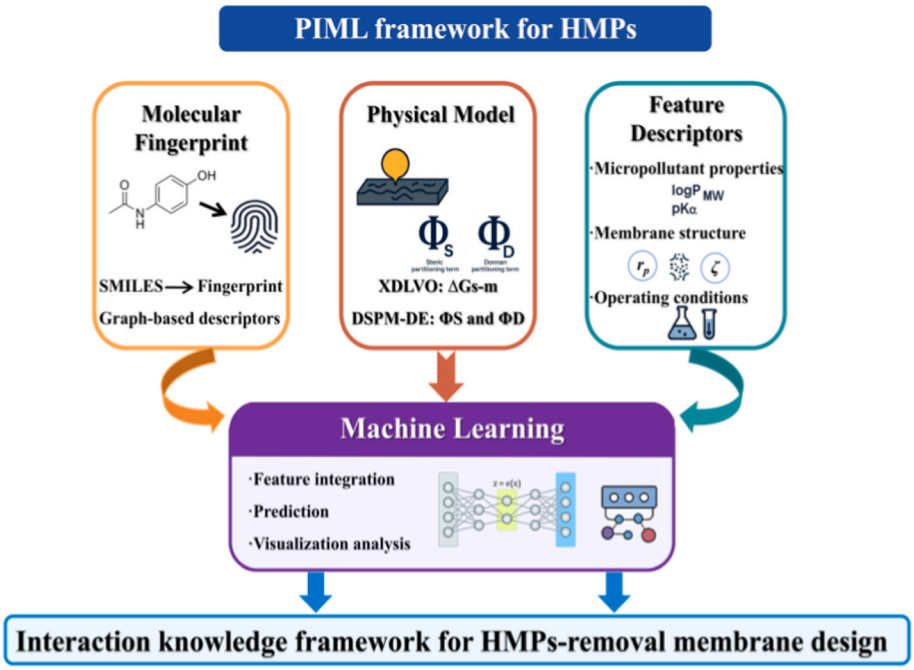


Fig. 1. | Overview of the PIML framework for HMPs.

subsequent analysis [23,24]. The distribution of these substructures were summarized in Fig. S1 (see in supplementary material).

In parallel, to mitigate feature redundancy and enhance model robustness, we implemented a systematic, algorithm-driven feature selection procedure. Absolute Pearson correlation coefficients were calculated among all numerical descriptors, and highly correlated feature pairs ($|r| > 0.7$) were identified as redundant [25]. A correlation network–based filtering algorithm was then applied to iteratively remove strongly correlated descriptors while preserving the key mechanistic parameters (Φ_S, ΔGs–m, and Φ_D). This objective, data-driven approach ensured that the final descriptor set was statistically independent and physically interpretable. The outcomes of this integrated selection process were illustrated in Fig. 2, which highlights the retained descriptors deemed both statistically independent and mechanistically informative.

To further validate descriptor independence and confirm that no essential structural information was lost, we extended this Pearson correlation analysis to include both the SMILES-derived substructure descriptors (Tier A) and the mechanistic quantitative descriptors (Tier B). The resulting correlation map (Supplementary Fig. S2) demonstrated that almost all descriptor pairs exhibit weak-to-moderate correlations, with only one chemically intuitive high-correlation pair (–CHO vs. C═O, $|r| > 0.7$). Moreover, correlations between Tier A and Tier B descriptors were uniformly low, confirming that the two descriptor tiers encode complementary and nonredundant information. This cross-tier validation further supports the robustness of the hybrid feature

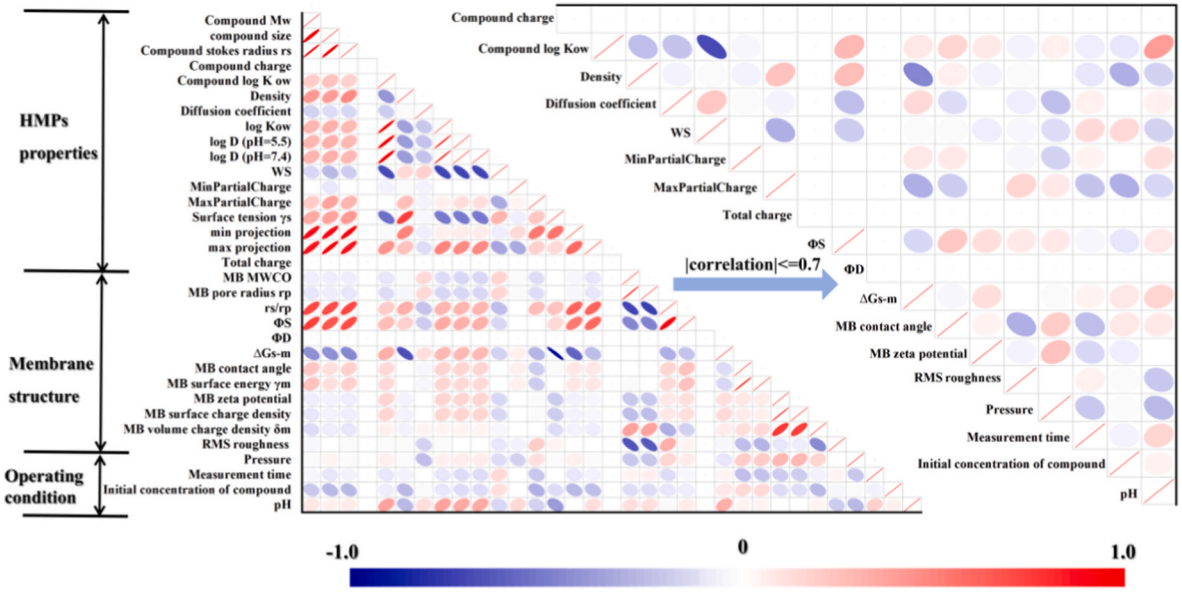


Fig. 2. | Feature Selection Process for Model Robustness Enhancement. This Figure illustrates the integrated feature selection process, combining Pearson correlation analysis and expert knowledge evaluation. Features with strong linear correlations ($|r| > 0.7$) were flagged as redundant, and subsequent expert assessment ensured that only the most relevant and independent descriptors were retained for further modeling.

framework and reinforces the interpretability and statistical soundness of the final model inputs.

This combined strategy—rooted in data-driven filtering and domain-guided selection—is consistent with best practices in ML, ensuring that the retained features contribute meaningfully to model accuracy and explainability.

2.3. Selection and implementation of ML for rejection prediction

To comprehensively explore predictive modeling of solute rejection, we implemented six machine learning algorithms representing two distinct modeling paradigms: tree-based ensemble models (RF, XGBoost, LightGBM) and neural network architectures (ANN, CNN, Transformer). All models were trained on the same curated molecular and membrane descriptor set, using consistent training–test splits to ensure fair comparison. Specifically, the dataset was randomly partitioned into a training set (80 % of the samples) and a test set (20 % of the samples), with stratification disabled to preserve the natural distribution of the target variable. Predictive performance quantified by the coefficient of determination (R^2), root mean square error (RMSE), and mean absolute error (MAE). This unified workflow was designed to identify high-performing models for subsequent interpretability analysis, while minimizing the influence of stochastic variability [26].

For the tree-based ensemble models, we applied systematic hyperparameter optimization to maximize generalization performance. The training pipeline employed GridSearchCV (scoring = 'r^2', cv = 5, n_jobs = −1) to exhaustively search combinations of key hyperparameters [27], including the number of estimators, maximum tree depth, learning rate (for XGBoost and LightGBM), and feature subsampling rates. Following grid search, parameters were manually fine-tuned to refine validation performance. The final hyperparameter configurations for each ensemble model were provided in Table 1. This two-stage optimization ensured that each algorithm was evaluated under its optimal configuration while maintaining comparability across models.

For the neural network models, architectures were implemented in PyTorch and trained using the Adam optimizer (learning rate = 1×10^{-3}) with mean squared error loss. The ANN comprised an input layer matching the descriptor dimension, two fully connected hidden layers with ReLU activation, and a linear output layer, trained for 1000 epochs. The CNN consisted of two one-dimensional convolutional layers (16 and 32 filters, kernel size = 3, padding = 1) with ReLU activation, followed by max pooling (pool size = 2) and two fully connected layers (64 and 1 units), also trained for 1000 epochs. The Transformer embedded descriptors into a 64-dimensional latent space via a linear projection, processed them through a single encoder layer (nhead = 4, feedforward dimension = 128, dropout = 0.1), and applied mean pooling before a regression output layer; this model was similarly trained for 1000 epochs. The best-performing checkpoint for each neural network was selected based on validation loss.

The use of multiple, diverse algorithms was intended to ensure that subsequent feature-importance analyses would not be biased by the idiosyncrasies of a single model [28]. Although the downstream interpretability stage would ultimately focus on one representative model, incorporating both ensemble-based and neural network-based predictors at the training stage provided a safeguard: if SHAP patterns were consistent across models, the mechanistic insights could be interpreted with higher confidence; if substantial divergence occurred, this would indicate model-dependent interpretations requiring further scrutiny. This multi-model design therefore not only enhanced predictive robustness but also laid a more reliable foundation for mechanistic exploration.

Table 1
Hyperparameter search ranges for tree-based ensemble models (RF, XGBoost, LightGBM).

Model	Hyperparameters	Range
RF	n_estimators	[100, 150, 200,300]
	max_depth	[1,10, 15, 20.30.50]
	min_samples_split	[1–5,7–10]
	min_samples_leaf	[1,2,4,8,10]
XGBoost	alpha	[0.1,1,10,20,30,40]
	max depth	[1–10]
	learning_rate	[0.01,0.1,0.2,0.3]
	n_estimators	[100, 200, 500]
	colsample_bytree	[0.1,0.2,0.3,0.4,0.5,0.6,0.7,0.8,0.9]
LightGBM	alpha	[0.1,1,10,20,30,40]
	max_depth	[1–10]
	learning_rate	[0.01,0.1,0.2,0.3]
	n_estimators	[100, 200, 500]
	colsample_bytree	[0.1,0.2,0.3,0.4,0.5,0.6,0.7,0.8,0.9]

2.4. Model interpretation techniques

2.4.1. SHAP-based feature importance analysis

To uncover the underlying mechanisms behind solute rejection and to enhance model transparency, we employed SHAP as a post hoc interpretation tool. SHAP is a game-theoretic approach that attributes the contribution of each feature to the model's output by computing Shapley values based on coalitional game theory [29]. This technique has been widely recognized for its consistency and local accuracy in interpreting complex ML models, particularly in fields such as cheminformatics and environmental modeling [30,31].

In this study, SHAP was applied to the trained model to quantify the influence of each input feature on the predicted rejection values. By aggregating the absolute SHAP values across the dataset, we obtained a global ranking of feature importance, enabling us to pinpoint which molecular or tri-domain descriptors had the most pronounced effect. Furthermore, SHAP summary plots were used to visualize both the magnitude and directionality of feature impacts, offering insights into how specific attributes promote or hinder rejection.

This interpretability framework not only aids in validating the physical relevance of selected features but also provides mechanistic insights into solute–membrane interactions, thereby serving as a bridge between data-driven predictions and domain knowledge.

2.4.2. PDP visualization

To enhance the interpretability of the machine learning models, we employed PDP analysis to explore the marginal effects of individual features on model predictions. PDPs estimate the expected output of a model as a function of one or two selected input variables, while averaging out the influence of all other variables in the dataset [32]. This approach is particularly useful for visualizing nonlinear relationships and interaction effects in complex models such as gradient boosting frameworks.

In this study, PDPs were generated for key descriptors identified by feature importance metrics to qualitatively assess their contributions to the predicted rejection values. The plots were constructed using the trained models and implemented using standard interpretation libraries. This technique complements SHAP analysis by offering a global perspective on feature-response relationships, thereby supporting mechanism-aware evaluation of model behavior [33].

2.5. MD simulations

All MD simulations were performed using the GROMACS 5.0.7 simulation package [34]. To construct the polyamide membranes, we employed the PXLink algorithm [35], which enableed dynamic covalent bond formation between trimesoyl chloride (TMC) and *m*-phenylenediamine (MPD) monomers. This approach allowed for the generation of crosslinked membranes with target DC of 20 % and 80 %, respectively. The resulting structural models—presented in Fig. S3—were finalized by terminating further bond formation once the desired DC was reached, thereby preserving the membrane's final configuration. The DC

was subsequently calculated using the following expression.

$$DC(\%) = 100 \times \frac{N_{\text{fully crosslinked TMC}}}{N_{\text{fully crosslinked TMC}} + N_{\text{partially crosslinked TMC}}}$$

Molecular parameters for the system were adopted from the CHARMM36 force field, which has been extensively validated for MPD–TMC polyamide systems [36–38]. Water molecules were modeled using the TIP3P model. All simulations employed a leapfrog integrator with a 2 fs time step, and system temperature was maintained at 300 K using the velocity-rescaling (V-rescale) thermostat. Long-range electrostatic interactions were computed using the Particle Mesh Ewald (PME) method with a real-space cutoff of 1.2 nm, while Lennard-Jones (LJ) interactions were truncated at the same distance. Periodic boundary conditions were applied uniformly in all spatial directions.

To examine how membrane crosslinking density and solute polarity influence interfacial interaction strength, we conducted simulations using membranes with 20 % and 80 % DC. It is noteworthy that membranes with lower crosslinking density tend to exhibit a higher surface concentration of carboxylic acid groups, attributed to the incomplete conversion of TMC. To mimic the initial solute–membrane approach observed during rejection processes, small organic molecules bearing either one or three hydroxyl groups were initially positioned approximately 0.5 nm above the membrane surface.

Each simulation system underwent a 500 ps equilibration phase in the canonical (NVT) ensemble, followed by a 10 ns production run under the same conditions. In all cases, spontaneous adsorption of solute molecules onto the membrane surface was observed within the first 2–4 ns, underscoring the strength of interfacial interactions. Subsequent post-processing was conducted using GROMACS built-in analysis tools to quantify nonbonded interaction energies—specifically, the short-range Lennard-Jones (LJ-SR)—between solutes and the polyamide matrix. These values were used to approximate ΔGs-m, which served as a quantitative metric for comparing interaction strength across different membrane and solute configurations.

3. Result and discussion

3.1. Evaluation and selection of predictive models

To comprehensively assess predictive performance and ensure the robustness of feature attribution, we benchmarked six ML algorithms representing both tree-based and neural network paradigms: RF, XGBoost, LightGBM, CNN, Transformer, and ANN. These models were selected for their proven capability in handling high-dimensional, nonlinear relationships, and their complementary algorithmic mechanisms—ranging from ensemble decision trees to deep learning architectures—thus providing a broad methodological spectrum for evaluation [39,40].

Optimal hyperparameters for tree-based models were obtained through grid search with subsequent manual fine-tuning, as summarized in Supplementary Table 2. Model performance was evaluated using three standard regression metrics—R^2, MAE, and RMSE —on both training and independent test sets [41,42]. As summarized in Table 2, XGBoost achieved the highest R^2 and lowest error values across datasets, followed closely by LightGBM and the Transformer. Neural network models, while competitive in training performance, exhibited slightly higher test errors, suggesting a greater tendency toward overfitting under the given dataset size and feature distribution [43]. Box plots of R^2 from 10 repeated train-test splits (Fig. 3) further confirmed that XGBoost offered both superior median accuracy and the lowest variance among all models, indicating enhanced generalization and stability. To further probe potential systematic biases that may be masked by global metrics, we inspected prediction quality across models and performed residual analyses. Fig. S4 presented true-versus-predicted rejection values for all six models on training and independent test sets: XGBoost showed the tightest agreement and the fewest large deviations. Based on this, we further examined XGBoost residuals: Fig. S5 showed the histogram of relative errors (Predicted-Experimental, %), revealing that the majority of test-set residuals lay within ±5 %, while only a small fraction fell in the ±5–10 % ranges and very few exceeded ±10 %. These findings indicated that the model did not exhibit pronounced systematic bias across the examined data subsets, justifying its use for the SHAP/PDP interpretability work presented below.

Table 2
Performance comparison of RF and XGBoost models.

	Set	R^2	MAE	RMSE
RF	**Training**	0.951	4.074	6.722
	Testing	0.815	8.7289	13.771
XGBoost	**Training**	0.980	2.656	4.287
	Testing	0.908	5.954	9.701
LightGBM	**Training**	0.977	2.770	4.628
	Testing	0.871	6.976	11.480
CNN	**Training**	0.945	5.040	7.116
	Testing	0.876	7.828	11.280
Transformer	**Training**	0.964	3.941	5.767
	Testing	0.874	6.418	11.339
ANN	**Training**	0.925	4.688	8.299
	Testing	0.821	8.302	13.509

To connect predictive outcomes to mechanistic interpretation, SHAP feature attribution analyses were performed for all six models (Fig. S6). Notably, the top-ranked descriptor, ΔGs-m, consistently emerged as the most important predictor of solute rejection across all models, irrespective of ML model. While the ranking of secondary features (Φ_S, hydroxyl group count) showed minor variations, their SHAP dependence plots supported broadly similar directional trends, suggesting that the physical-chemical relationships uncovered were robust to model choice. This convergence alleviated concerns that interpretative conclusions could be artifacts of a single modeling framework [43,44].

Given its combination of highest predictive accuracy, stability across repeated experiments, and interpretability-friendly structure, XGBoost was selected as the primary model for subsequent detailed SHAP-based mechanistic analysis. This choice leveraged the algorithm's ability to balance predictive performance with transparent feature contribution profiling, while ensuring that the insights drawn were supported by a broad multi-model comparison.

3.2. Rejection mechanism of HMPs from PIML

3.2.1. Interpretable insights from SHAP-based feature attribution

Across all six predictive models, SHAP-based feature attribution supported broadly consistent importance rankings for the descriptors, with ΔGs–m persistently emerging as the most influential predictor of solute rejection. This convergence across tree-based and neural network architectures suggested that the identified key features were not artifacts of a particular modeling approach, thereby lending greater confidence to their mechanistic relevance. Based on this consistency, we selected the XGBoost model for detailed interpretability analysis, as it offered both high predictive accuracy and a clear feature-importance structure. Fig. 4a presented the ranked top ten descriptors from a representative XGBoost run, while Fig. 4b summarized the ranking stability across ten independent runs with varying random seeds.

A more focused analysis of the XGBoost model was provided in Fig. 4a, which ranked the top ten most impactful descriptors. These included not only physically-mechanism-based parameters—such as ΔGs-m and Φ_S —but also molecular structures like –OH group, reflecting the importance of physicochemical and structural determinants of separation performance. Recognizing that SHAP importance rankings may fluctuate under different random seeds or training splits, we conducted ten independent XGBoost runs with varying initializations [45]. The corresponding ranking distributions presented in Fig. 4b.

Among all features, ΔGs–m consistently ranked as the most important predictor of rejection, occupying the top position across all SHAP

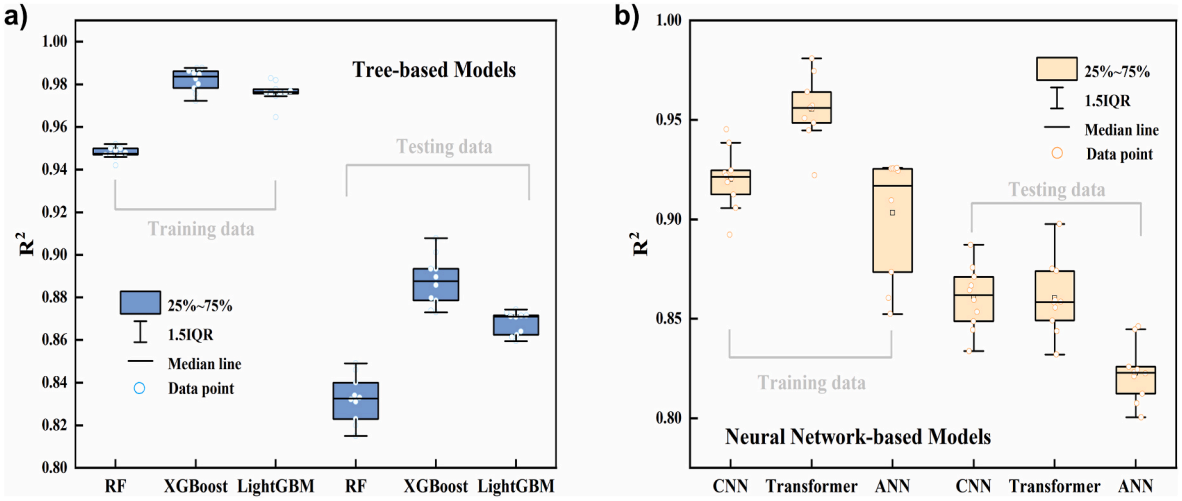


Fig. 3. | Distribution of R^2 Scores from Repeated Model Training. (a) Tree-based Models — Box plots showing R^2 distributions across 10 repeated experiments. among tree-based models, XGBoost achieves the highest median and the lowest variance, indicating superior robustness. **(b) Neural Network-based Models — Box plots showing R^2 distributions for neural network models.** Transformer outperforms other neural architectures.

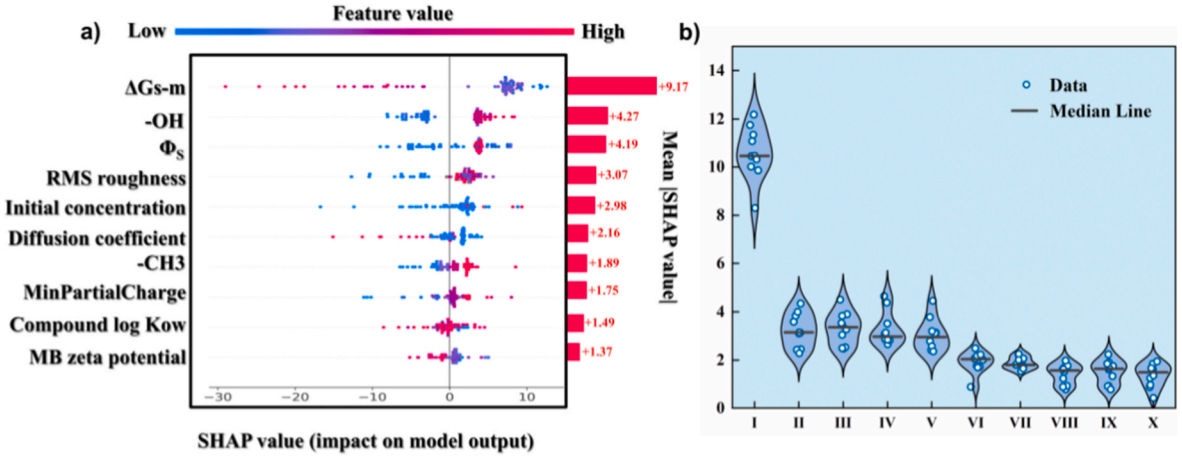


Fig. 4. | SHAP-Based Interpretability Analysis of the XGBoost Model.(a) Top 10 Features Ranked by SHAP Importance in XGBoost Model. This bar plot summarizes the ten most influential descriptors contributing to solute rejection, as determined by their mean absolute SHAP values, highlighting the relative impact of each feature on rejection.**(b) Variability of SHAP Feature Rankings Across 10 Independent Runs.** Roman numerals I–X correspond to the top 10 features in Fig. 4a. It illustrate the positional consistency of top-ranked features in repeated SHAP analyses, revealing that ΔGs-m maintains stable importance while other descriptors, such as Φ_S and –OH, show greater variability.

analyses. As shown in Fig. 4a, one representative run yielded a SHAP value of 10.47 for ΔGs–m, while its values across all predictions ranged from 8.30 to 12.18 (Fig. 4b), underscoring its dominant influence. Notably, SHAP dependence plots supported a clear monotonic trend: as ΔGs–m becomes more negative, the rejection markedly increases, indicating stronger solute-membrane affinity and consequently greater rejection.

In contrast, Φ_S and –OH, while subject to ranking fluctuations, consistently appeared among the top four features. In the specific case illustrated in Fig. 4a, Φ_S (SHAP value range: 2.29–4.35 in Fig. 4b) and –OH (SHAP value range: 2.48–4.51 in Fig. 4b) exhibited SHAP values of 4.35 and 3.28, respectively. This variability arose from both statistical and mechanistic factors. From the modeling perspective, ΦS and –OH displayed comparable average |SHAP| magnitudes (≈3–4), substantially lower than that of ΔGs–m (≈10). When multiple descriptors contributed similar levels of variance, small perturbations introduced by data resampling or tree-splitting thresholds can slightly redistribute feature attributions within the ensemble, yielding apparent rank shifts without compromising model stability. From a mechanistic standpoint, these fluctuations reflected the context-dependent interplay between steric and chemical effects that jointly govern solute rejection. The contribution of ΦS was most pronounced when solute size approacheed the membrane's sieving threshold, whereas its importance diminished when ΔGs–m became strongly negative and interaction energy dominated transport behavior. Conversely, the influence of –OH depended on solute polarity and hydrogen-bonding capacity: OH–rich solutes interacted more strongly with polar or hydrophilic membrane surfaces, enhancing retention through interfacial association or restricted diffusion. Variability in the polarity distribution of HMPs across training subsets therefore leaded to moderate fluctuations in the relative importance of –OH. Despite their secondary importance, both features contributed non-negligibly to rejection behavior. In particular, the –OH group demonstrated a clear functional effect: HMPs with more hydroxyl substituents tended to exhibit higher rejection by membrane.

Importantly, our SHAP-based interpretability analysis suggested a possible mechanistic deviation from conventional assumptions in nanofiltration science. While size exclusion—quantified here as the steric hindrance parameter Φ_S—has long been considered the dominant

factor governing organic micropollutant rejection [46], our model consistently identified ΔGs-m as the most critical predictor across all training iterations. This shift underscored a pivotal insight: in the case of HMPs, solute-membrane interaction energies rather than purely geometric constraints predominantly determine membrane rejection. Derived from the extended DLVO framework, ΔGs-m captured the total free energy landscape governing solute–membrane affinity, encompassing van der Waals forces, electrostatic interactions [47,48].

More negative ΔGs–m values were strongly associated with higher rejection rates, indicating an interaction-energy–governed exclusion trend for HMPs (Fig. 4). A key explanation was that more negative ΔGs–m values indicate stronger affinity toward the membrane interface. While one might expected such affinity to facilitate permeation, our findings suggested the opposite: solutes exhibiting high interfacial adhesion tended to be retained at the membrane surface without crossing into or through membrane pores. This mechanism was consistent with the enhancing solute–membrane affinity could hinder transmembrane diffusion in hydrophilic systems [49].

Although Φ_S emerges as one of the more influential descriptors, its SHAP value was similar as that of –OH groups, RMS roughness, and initial compound concentration. One possible explanation was that steric hindrance alone could not fully account for the nuanced interplay between membrane structure and solute rejection under conditions of strong interfacial interactions.

We specifically highlighted the role of –OH groups due to their unique structural and mechanistic implications. The substantial contribution of polar functional groups, –OH, further enhanced our understanding on molecular-scale separation mechanisms. SHAP showed that HMPs with a greater number of hydroxyl groups generally exhibited higher rejection rates. The enhanced polarity of HMPs due to –OH groups facilitated stronger solute–membrane interactions through hydrogen bonding and dipole–dipole interactions, especially when the membrane surface also possessed polar functional groups. This increased affinity resulted in greater adsorption and higher rejection efficiency. However, it was worth noting that while –OH groups contribute to solute–membrane affinity, they were not the sole determinant of the interaction free energy. The latter was a comprehensive descriptor influenced by multiple molecular features, including size, shape, and charge distribution. Therefore, a higher number of –OH groups did not always translate directly into a stronger ΔGs–m. From a chemical standpoint, hydroxylated compounds tended to form stable hydration shells and exhibited increased interfacial interaction with polar membrane surfaces. Such affinity did not necessarily facilitate membrane passage; instead, it could result in stronger surface association and reduced transmembrane mobility. These findings were consistent with prior theoretical and simulation work in solvation dynamics and molecular-scale adhesion [50,51], where hydration-mediated interactions created interfacial barriers that favor surface retention over permeation.

3.2.2. Consistency and sensitivity assessment of surface tension (γ) estimation in the ΔGs-m

To ensure the physical reliability and numerical robustness of the key energetic descriptor ΔGs-m, we systematically evaluated the consistency and sensitivity of its dependence on the γ. As γ directly entered the XDLVO-based formulation of ΔGs-m, uncertainty in its empirical estimation could, in principle, affect the derived interaction energetics and model interpretation.

To reduce this uncertainty, a machine learning regression model was developed to predict γ using experimentally reported physicochemical properties of HMPs, including molecular weight, polarity, dipole moment, and functional group composition. The predicted γ values showed strong agreement with reported experimental data (Fig. S7), confirming the reliability of this approach. These predicted values were then used to refine the computation of ΔGs-m across the dataset, which was subsequently employed for model retraining. Updated SHAP and PDP analyses (Figs. S8 and S9) supported that the refined γ estimation did not materially alter the relative importance or interpretability of ΔGs-m, reaffirming its dominant role in governing solute rejection.

To further test robustness, we performed a quantitative sensitivity analysis by introducing random perturbations of γ within ±10 % for all solutes, reflecting typical experimental or empirical uncertainty ranges. Ten perturbed datasets were generated, and the PIML model was retrained under identical hyperparameter settings. The SHAP results were shown in Fig. S10. The resulting statistical summary (Table S3) showed that the model's R^2 fluctuated within 0.02, with MAE and RMSE variations below 8 %. SHAP analysis further confirmed that ΔGs-m remained the top-ranked feature in all retrained models (Fig. 5).

These findings confirmed that both the magnitude and mechanistic interpretation of ΔGs-m were stable under realistic γ perturbations. The combined resulted from regression-based refinement, consistency checks, and sensitivity tests demonstrated that ΔGs-m was a physically meaningful and statistically robust descriptor. The analysis also reinforced that γ, while influential, did not introduce significant uncertainty into the mechanistic insights derived from the PIML framework.

3.2.3. Quantifying feature effects via PDP analysis

Building upon the interpretability insights provided by SHAP analysis, we further employed PDP to quantify the global impact of ΔGs–m and –OH on rejection, offering a more intuitive depiction of how interaction energy and molecular structure respectively influence membrane rejection. PDP analysis captured the average predictive response over the entire dataset, offering a more intuitive visualization of the dominant factors shaping membrane performance.

As a preliminary step, the SHAP dependence in Fig. S11 (see in supplementary material) offered valuable insight into feature heterogeneity and non-linear effects across individual samples [52]. Notably, ΔGs–m exhibited a clearly monotonic relationship with SHAP values, indicating a consistent and dominant role in rejection prediction. The PDP results (Fig. 6) further elucidated these trends. For ΔGs–m (Fig. 6a), the marginal effect is strongly and negatively correlated with rejection rate. A distinct inflection behavior is observed within a critical transition region of ΔGs–m, below which the rejection sharply increases. This threshold was derived from the present system—polyamide membranes and a comprehensive dataset of HMPs encompassing a wide range of molecular sizes, polarities, and functional groups. Although the value of ΔGs-m ≈ -1.0 J m^{-2} reflected the energetic landscape specific to this system, the underlying mechanistic implication was expected to be broadly applicable. This value served as a critical energetic threshold, separating two interfacial regimes: When ΔGs-m > -1.0 J m^{-2} (less negative), solute–membrane interactions were relatively weak. Solutes may transiently adsorb on the membrane surface but could readily desorb and permeate through the membrane, resulting in lower rejection; When $\Delta G_{s-m} < -1.0$ J m^{-2} (more negative), the interaction became sufficiently strong to promote adsorption of the solute at the membrane interface. This prolonged residence time increased the likelihood of interfacial trapping or exclusion, thereby elevating the rejection rate. This threshold behavior aligned with the transition from weak to strong solute–membrane affinity and conformed to the phenomenon of HMPs were increasingly rejected. While the precise threshold value might vary depending on membrane chemistry, solvent conditions, or solute type, the qualitative energetic transition delineating these two transport regimes was mechanistically robust and physically meaningful. It was consistent with previous studies highlighting the role of interaction energy and hydration forces as energetic barriers in selective membrane transport [53,54]. This finding also provided new insight into the dual role of adsorption in nanofiltration processes. While adsorption had traditionally been regarded as a precursor to membrane contamination and flux decline [55], our results supported a distinct, energetically governed regime in which interfacial adsorption contributes positively to solute rejection. Specifically, when ΔGs-m became sufficiently negative, solutes were trapped at the membrane–water

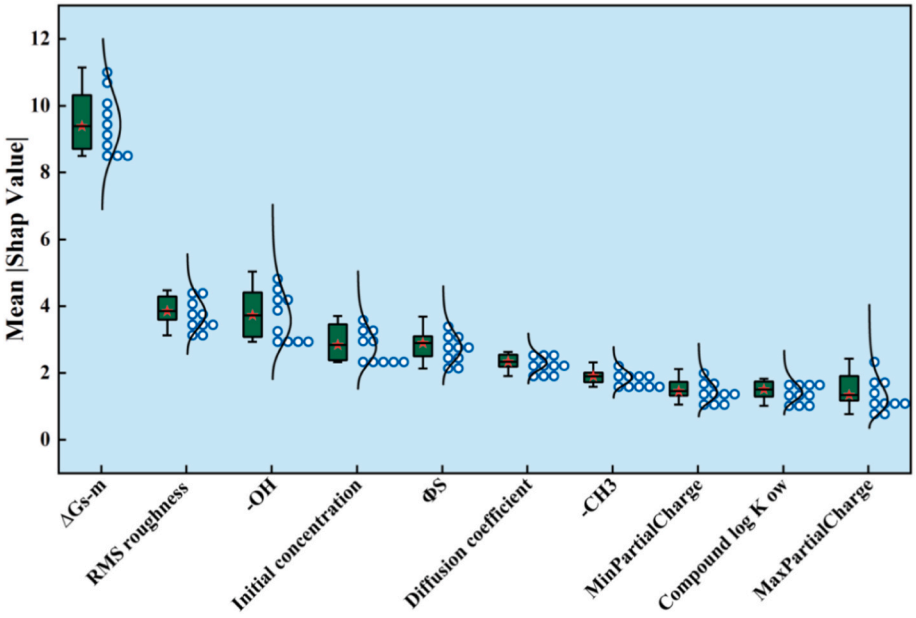


Fig. 5. | Sensitivity analysis of γ and stability of SHAP-based feature importance rankings.

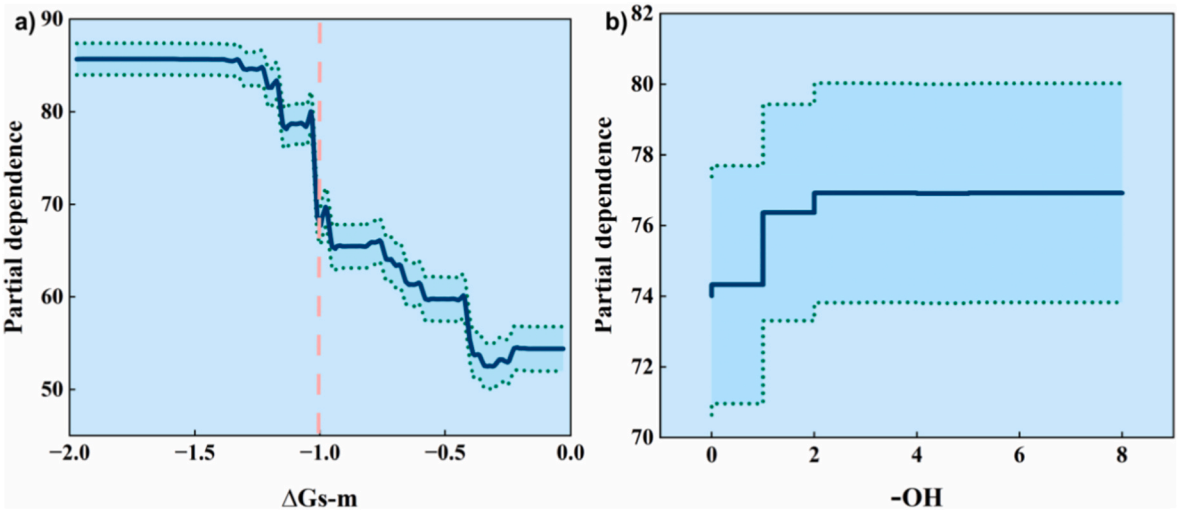


Fig. 6. | Partial dependence analysis of key features influencing rejection. (a) PDP of Δ Gs-m. A distinct nonlinear relationship between solute–membrane interaction free energy and rejection. A critical threshold appears near −1.0, beyond which rejection increases sharply. **(b) PDP of number of –OH group.** A saturating effect of hydroxyl functional groups on rejection. The contribution increases rapidly up to two –OH groups and then plateaus, suggesting diminishing marginal impact due to limited hydrogen bonding or hydration interactions at the membrane interface.

interface by interaction energies, rather than penetrating into or blocking the membrane pores. This transient, interfacial-scale adsorption increased the residence time of solutes at the surface and reduces their transmembrane diffusion probability, effectively enhancing rejection without inducing fouling [56]. Hence, the adsorption-governed rejection observed here should be interpreted as a transient interfacial regulation rather than a fouling-related accumulation process. In this regime, solute molecules were momentarily captured at the membrane–water interface through favorable energetic interactions, which hindered their passage without obstructing water transport.

Similarly, the PDP curve for –OH group (Fig. 6b) highlighted a saturating effect: rejection efficiency rose sharply with the presence of up to two hydroxyl groups, after which the incremental impact plateaus. This plateauing effect likely reflected the finite number of hydrogen bonding sites available on the membrane surface, as well as the reduced marginal contribution of excessive polar functionalities due to interfacial polarity mismatch. Similar saturation effects had been observed in solvation shell studies, where the hydration environment became structurally complete beyond a critical number of hydrogen-bond donors [57]. This observation reinforced the notion that functionalization of –OH group primarily enhanced rejection through polar interactions, but its impact was constrained by the cooperative limits of hydrogen bonding at the interface. To dissect the interactive effects between key descriptors, bivariate PDPs were constructed in Fig. S12 (see in supplementary material). The plot for ΔGs–m and –OH (Fig. S12a) supported discrete response regions defined by abrupt shifts in rejection probability, particularly when ΔGs–m fell below −1.0 J m^{-2}. This observation confirmed the critical nature of this interfacial energy threshold and suggested that polar functional groups such as –OH group acted synergistically with weak solute–membrane interaction to enhance exclusion, possibly by destabilizing transient binding events through hydration-mediated repulsion. In comparison, the joint

PDP of ΔGs–m and Φ_S (Fig. S12b) demonstrated that while Φ_S only significantly influenced rejection within a specific regime – namely, when solute size was near the membrane's size cutoff point. This was particularly notable given the traditional emphasis on steric hindrance as a dominant mechanism in nanofiltration [58].

It uncovered a rejection mechanism in which solute–membrane interaction energetics, as quantified by ΔGs–m, served as the primary determinant of performance, while features such as –OH groups and Φ_S acted as modulators whose influence depended on the specific molecular and membrane context. From a membrane design perspective, this highlighted the importance of the membrane surface property—such as modifying membrane polarity, hydration propensity, or surface energy—to selectively weaken unfavorable solute interactions and maximize rejection efficiency. This mechanistic framework aligned the rejection performance was governed by interaction rather than solely by steric hindrance. Our PDP-based analysis further supported the importance ΔGs–m as both a predictive and mechanistic descriptor of membrane performance. By bridging model interpretability with physicochemical understanding, these findings provided a rational basis for designing next-generation membranes tailored for the efficient removal of HMPs.

3.2.4. *Structure-based prediction and mechanistic interpretation of HMP rejection*

Building upon the mechanistic understanding that ΔGs–m serves as a key energetic descriptor governing solute rejection, we further explored how structure-informed modeling could be used to predict and rationalize rejection behavior. Structure-based quantitative structure–activity relationship (QSAR) models have shown promise in predicting solute rejection by encoding molecular features into data-driven frameworks. In this context, the PIML framework advances beyond purely empirical correlations by incorporating physically interpretable structural descriptors—ΔGs–m and Φ_S—that capture the energetic and steric contributions of solute–membrane interactions.

In the context of HMPs, the two descriptors— ΔGs-m and Φ_S—represented controllable structural features that governed the physical rejection mechanism. Both descriptors emerged as among the most influential factors in SHAP analysis, reflecting their central role in determining solute transport behavior.To further elucidate this structure–performance relationship, four representative HMPs were selected for detailed examination based on the bivariate PDPs results: Phenacetin ($\Delta G_{s-m} > -1$ J m^{-2}, $\Phi_S > 0.25$; rejection = 56.00 %), Benzonitrile ($\Delta G_{s-m} > -1$ J m^{-2}, $\Phi_S < 0.25$; rejection = 40.10 %), Hydrocortisone ($\Delta G_{s-m} < -1$ J m^{-2}, $\Phi_S < 0.25$; rejection = 81.78 %) and Caffeine ($\Delta G_{s-m} < -1$ J m^{-2}, $\Phi_S > 0.25$; rejection = 95.51 %).

The corresponding structure–property distributions were illustrated in Fig. 7, where predicted and experimental values exhibited excellent agreement, confirming the accuracy of PIML in capturing structure-dependent rejection trends. The analysis supported two mechanistically distinct regimes. When $\Delta G_{s-m} > -1.1$ J m^{-2}, rejection was dominated by steric constraints and remains moderate. In contrast,

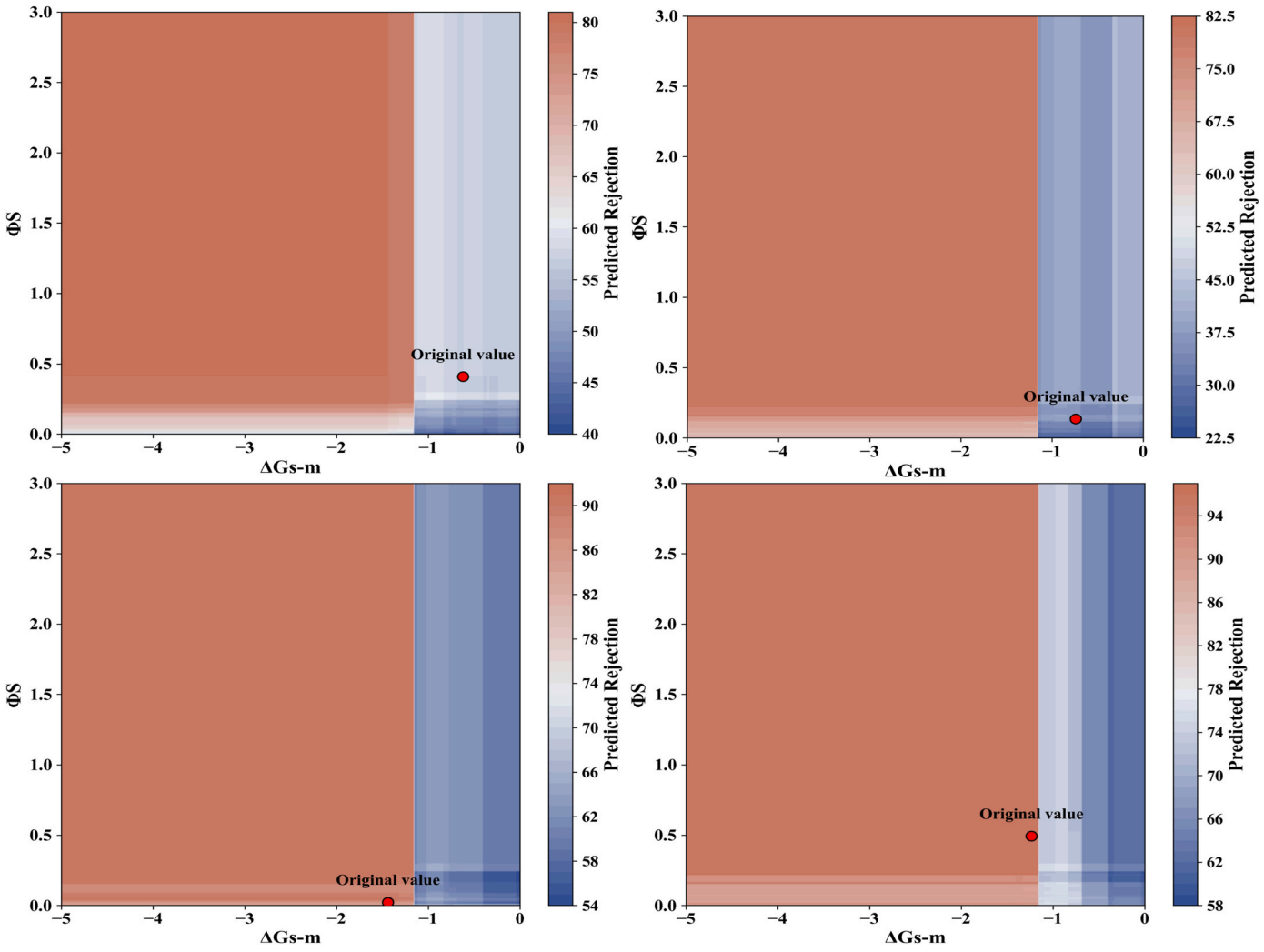


Fig. 7. | Structure–property relationship map illustrating how ΔGs-m and ΦS jointly govern the rejection behavior predicted by PIML.

when $\Delta G_{s-m} < -1.1$ J m^{-2} and $\Phi_S > 0.25$ (rp > 2 rs), rejection efficiency markedly increased, indicating that strong solute–membrane interaction energetics synergize with size exclusion to suppress transmembrane transport.

These findings demonstrated that PIML not only reproduced the predictive capability of structure-based DL-QSAR approaches but also extended it by embedding physically interpretable descriptors that directly correspond to molecular and membrane structures. The framework thus enabled the rational tuning of membrane surface chemistry and pore structure to achieve targeted removal of HMPs through structure–mechanism coupling.

3.3. Solute-membrane interaction via MD simulations

To substantiate the mechanistic insights derived from the PIML framework—particularly the role of ΔG_{s-m} in governing the rejection behavior of HMPs—we conducted atomistic MD simulations on a set of representative solute–membrane systems. Two model HMPs, Acetaminophen (bearing a single hydroxyl group) and Dexamethasone (with three hydroxyl groups), were selected to systematically probe the influence of solute polarity. The molecular structures of the two solutes are provided in Supplementary Fig. S13. The selection of these molecules was motivated by their distinct chemical characteristics: acetaminophen is a relatively small, hydrophilic molecule with limited polar functionality, while dexamethasone presents a larger, more rigid scaffold with multiple hydroxyl groups, thereby representing increased polarity and steric complexity. These contrasting molecular features made them ideal candidates for capturing the polarity–rejection relationship across structurally diverse solutes.

To explore how membrane structure affects this relationship, the two solutes were simulated in contact with a polyamide membrane possessing a low DC (20 %), representing a flexible and chemically heterogeneous surface. Fig. 8a illustrateed the initial adsorption configurations of these solute–membrane systems, including acetaminophen (with one –OH group) and dexamethasone (with three –OH groups) on the 20 % crosslinked membrane, as well as dexamethasone on the 80 % crosslinked membrane. This visual comparison underscored the differences in solute–membrane proximity and interfacial orientation at the onset of interaction. Additionally, to isolate the effect of membrane rigidity and network density, a separate simulation was conducted for dexamethasone interacting with a high DC (80 %) membrane. In all cases, the van der Waals (vdW) interaction energy between the solute and the membrane matrix was calculated and used as a proxy for ΔG_{s-m}. It should be noted that the computed van der Waals and electrostatic interaction energies represented the static solute–membrane affinity, rather than the complete free-energy barrier associated with molecular permeation. These adsorption energies served as a computationally tractable and physically meaningful proxy to assess the strength of solute–membrane interactions [59,60]. While adsorption energy quantified the interfacial affinity rather than the complete permeation barrier, previous theoretical and simulation studies have demonstrated a strong thermodynamic coupling between these two quantities, where deeper adsorption correspond to higher free-energy barriers [61,62]. These simulations enabled a molecular-level evaluation of how solute polarity and membrane microstructure jointly shape the interfacial interaction landscape, thereby influencing solute retention.

As shown in Fig. 8b, with the number increase of hydroxyl groups from one (I) to three (II), the vdW interaction energy increased from −79.67 to −148.67 kJ/mol in the system with the 20 % DC membrane. Notably, the molecule containing three –OH groups exhibited a markedly tighter adsorption configuration at the membrane interface, as evidenced by its closer proximity and more extensive contact area with the polymer surface. This enhanced interfacial affinity was in agreement with the prediction results of PIML framework: more negative ΔG_{s-m} values corresponded to stronger solute–membrane interactions and, consequently, higher rejection rate, as previously established through SHAP and PDP analyses.

When the same three-hydroxyl solute was introduced to a more densely crosslinked (80 %) membrane (III), the interaction energy decreased to −105.15 kJ/mol. It could be attributed to the reduced availability of interfacial polar sites. As the degree of membrane

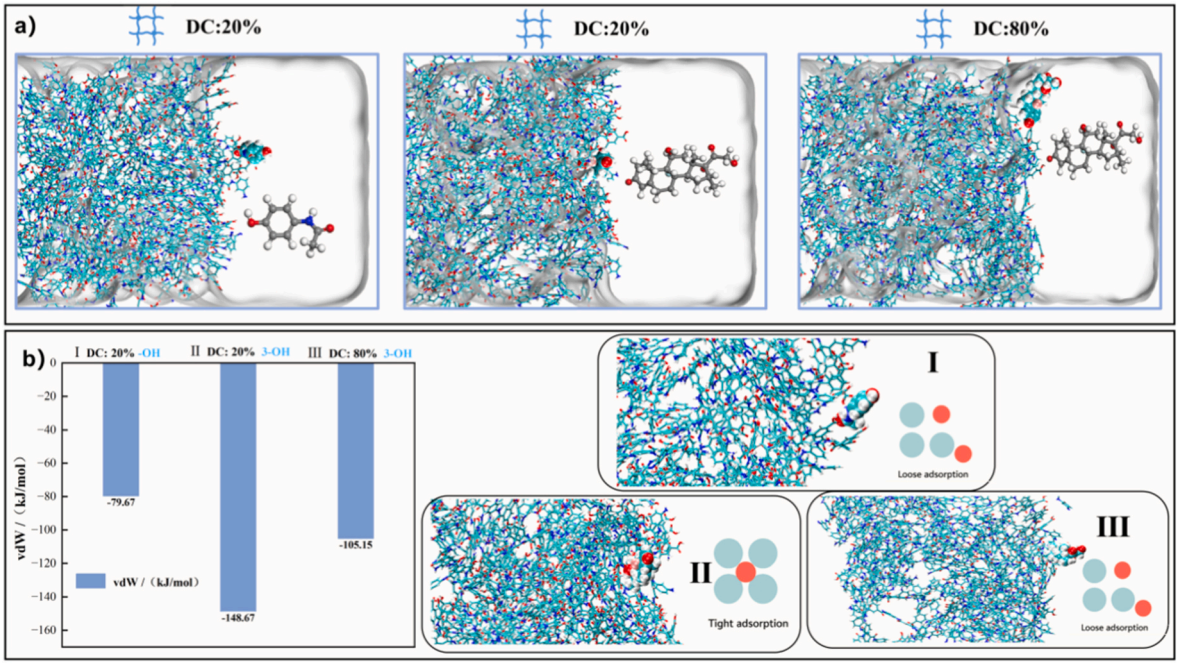


Fig. 8. | MD simulations uncover polarity–structure synergy in governing solute–membrane interactions. (a) Representative MD snapshots of early-stage solute adsorption under three structural regimes. (b) Interaction Behavior of Hydroxylated Solutes with Crosslinked Membranes of Varying Densities. Interaction energy between a mono-hydroxylated solute and a 20 % crosslinked membrane (I). Enhanced binding of a tri-hydroxylated solute to a 20 % crosslinked membrane (II). Attenuated interaction of the same solute with a highly crosslinked (80 %) membrane (III).

crosslinking increases, the number of available TMC molecules participating in the reaction decreases, leading to a significantly lower density of –COOH groups on the membrane surface. Thus, the membrane surface also could affect the interaction between solute and membrane. In the view of ΔGs–m as a pivotal energetic descriptor underlying solute rejection, rational tuning of membrane microstructure such as modulating crosslinking density could offer a viable strategy to regulate ΔGs–m and, in turn, tailor rejection performance for HMPs. To further dissect the energetic origin of ΔGs–m, we decomposed the total solute–membrane interaction energy into its short-range Coulombic (Coul-SR) and Lennard–Jones (LJ-SR) components. This decomposition provided a more mechanistic understanding of the respective roles of electrostatic and van der Waals interactions. The results (Supplementary Fig. S14) supported that LJ-SR consistently dominates the attractive interactions across all systems, particularly for larger and more hydroxylated solutes such as dexamethasone, highlighting the importance of surface contact area and geometric complementarity in governing solute–membrane affinity. In contrast, Coul-SR contributions, although smaller in magnitude, increased modestly with solute polarity, reflecting enhanced hydrogen-bond or dipole–dipole interactions at the interface. Increasing membrane crosslinking from 20 % to 80 % weakens both Coul-SR and LJ-SR terms, consistent with reduced polar-site accessibility and restricted molecular accommodation in the denser network. This component-resolved analysis confirmed that ΔGs–m effectively captured the synergistic effects of van der Waals and electrostatic interactions, providing a physically grounded basis for tuning membrane microstructure to control solute rejection.

3.4. Limitations of the study

While the PIML framework provided valuable mechanistic insights into the rejection of HMPs by polyamide membranes, several limitations should be acknowledged to ensure balanced interpretation.

(1) Dataset limitations and strategies to enhance PIML generalizability.

The current PIML model is based on a curated dataset of 353 experimental points covering polyamide-based nanofiltration and reverse osmosis membranes and diverse solute properties. While sufficient to capture key solute-membrane interactions within this domain, the dataset remains limited relative to the full chemical and structural space of polymeric membranes and organic contaminants. This constrains the model's ability to generalize to novel solutes or membrane chemistries. Future work will focus on expanding the dataset through high-throughput experiments, simulation-derived descriptors, and literature-mined data, enabling broader coverage of solute-membrane interactions and enhancing the model's predictive power, transferability, and mechanistic interpretability across diverse separation systems.

(2) Simplifications and Limitations of the XDLVO-Based Approach.

The physicochemical descriptor ΔGs-m was calculated based on the XDLVO theory using the Lifshitz–van der Waals/acid–base approach to quantify solute–membrane hydrophobic interactions. While this method provided a tractable and physically meaningful estimation of interfacial energetics, it assumed idealized surface homogeneity, equilibrium interfacial conditions, and additive energetic contributions. Such approximations may have neglected the effects of hydration layers, conformational flexibility, and specific chemical binding that occurred at realistic membrane–solute interfaces. Moreover, the empirical fitting used to estimate surface tension components introduced additional uncertainty, particularly for polar HMPs whose acid–base components were not directly measurable. Future work incorporating ab initio or reactive molecular dynamics calculations could provide a more quantitative of ΔGs-m.

(3) Lack of Potential of Mean Force (PMF) for permeation barrier quantification.

PMF calculations, which rigorously quantify the *trans*-membrane free-energy barrier, were not performed in this study due to their prohibitive computational cost in dense amorphous polyamide matrices. Pulling large solute molecules through highly crosslinked networks requires extensive sampling of polymer relaxation, solvent reorganization, and multiple possible permeation pathways. Within accessible MD timescales (<100 ns), such rare permeation events are difficult to converge and may distort the membrane configuration, leading to large uncertainties in the computed barrier. Therefore, we focused on adsorption-based interaction energies to represent the interfacial component of ΔGs-m, providing a computationally tractable yet physically meaningful measure of solute–membrane affinity. Although this approach does not capture the full permeation barrier, it effectively reflects the dominant energetic factors governing solute rejection. Future work will employ enhanced sampling methods (umbrella sampling or metadynamics) to compute PMF profiles and quantitatively link ΔGs-m with the permeation energy barriers.

Despite these limitations, the framework established a physically informed and interpretable foundation for integrating data-driven and mechanistic modeling approaches, providing a robust basis for future membrane design and process optimization.

4. Conclusion

This work developed the PIML framework for guiding the molecular design of membranes to reject HMPs. By integrating molecular fingerprints, physics-based interaction descriptors, and structural parameters, PIML provided a comprehensive representation of the factors governing HMP rejection. Surpassing classical steric hindrance, our analysis suggested the ΔGs–m as the primary determinant of rejection, with a critical threshold ($\Delta G_{s-m} \approx -1.0$ J m^{-2}) below which rejection increases sharply. Membrane chemical functionality, such as hydroxyl groups, was also shown to modulate rejection, highlighting the importance of molecular-scale interactions. MD simulations supported these findings at the molecular scale, revealing how solute–membrane interactions and functional group accessibility control adsorption and rejection. By uniting data-driven prediction with physical insights and molecular-scale mechanisms, this study provided mechanistic insights and practical guidance, where rejection was governed not only by pore structure but also by interfacial energetics and chemical functionality.

CRediT authorship contribution statement

Zihang Zhao: Writing – original draft, Visualization, Software, Methodology, Data curation. **Ming Wu:** Visualization, Validation. **Dan Lu:** Visualization, Methodology, Data curation. **Zhe Kong:** Visualization, Validation. **Zunzhong Ye:** Writing – original draft. **Lijun Liang:** Writing – review & editing, Writing – original draft, Formal analysis, Conceptualization. **Lin Zhang:** Writing – review & editing, Writing – original draft, Formal analysis, Conceptualization.

Code availability

The model files used in this work are accessible on GitHub (https://github.com/effortV/PIML-understanding-the-rejection-mechanism-of-HMPs).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.memsci.2025.124925.

Data availability

The data supporting the findings of this study are available in the paper and its Supplementary Information files.

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